NUCLEIC ACID METABOLISM

INTRODUCTION:

- Deoxyribonucleotides are required for DNA synthesis.
- Ribonucleotides are required for RNA synthesis.
- Biosynthesis of purine and pyrimidine nucleotides is essential for DNA replication (cell division) and growth of all types of mammalian cells, bacteria and virus. If the supply of nucleotides is blocked cell division (viral replication) and growth is halted. So, compounds, which can block nucleotide biosynthesis effectively halt growth of cells, bacteria and virus. Indeed, many anti-tumor, anti-bacterial and anti-viral agents currently used are inhibitors of nucleotide (nucleic acid) biosynthesis.

Purines and pyrimidines are dietary nonessential components. Dietary nucleic acids and nucleotides do not provide essential constituents for the biosynthesis of endogenous nucleic acids. Humans can synthesize purine and pyrimidine nucleotides de novo, i.e. from amphibolic intermediates.

BIOSYNTHESIS OF PURINE NUCLEOTIDES

- The two purine nucleotides of nucleic acids are:
    1) Adenosine monophosphate, AMP
    2) Guanosine monophosphate, GMP
- Purine nucleotides can be synthesized by two pathways:
    1) De novo pathway (New synthesis from amphibolic intermediates).
    2) Salvage pathway.

1) DE NOVO BIOSYNTHESIS OF PURINE NUCLEOTIDES:

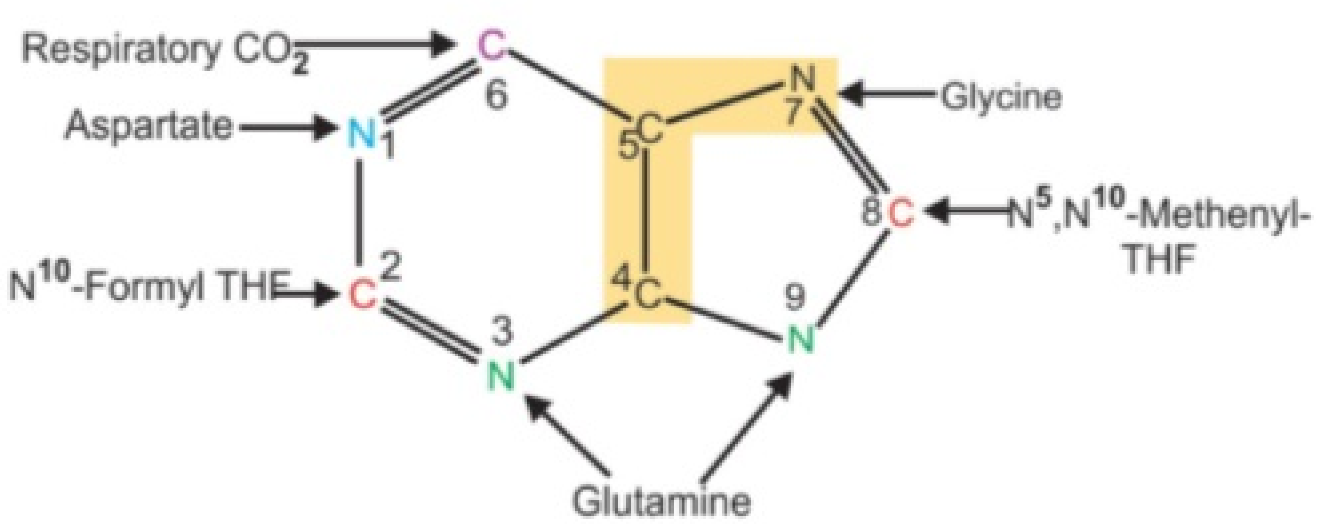


Figure 19.1: Source of carbon and nitrogen atoms in the purine ring

In de novo pathway, the purine ring formed from variety, of precursors (Figure 19.1) is assembled on ribose-5-phosphate.

PRECURSORS FOR THE DE NOVO SYNTHESIS OF PURINE:

- Glycine provides C_4, C_5 and N_7
- Aspartate provides N_1
- Glutamine provides N_3 and N_9
- Tetrahydrofolate derivatives furnish C_2 and C_8
- Carbon dioxide provides C_6.

MAJOR STEPS OF DE NOVO SYNTHESIS OF PURINE NUCLEOTIDES:

In Figure 19.2 different steps of de novo synthesis of purine nucleotides are numbered by Arabic numerals and Roman numerals designate structures in the pathway for convenience.

1) The biosynthesis of purine begins with ribose-5- phosphate (I), derived from pentose phosphate pathway, which is converted to phospho-ribosyl-pyrophosphate, PRPP (II) by the transfer of pyrophosphate group from ATP to C-1 of the ribose. This reaction is catalyzed by the enzyme **PRPP synthetase**.
2) PRPP is aminated by the addition of the amide group from **glutamine** to form amino sugar 5-phosphoribosylamine (III), The enzyme that catalyzes the transfer of the amide nitrogen is called **PRPP-amidotransferase**. The amino nitrogen of phosphoribosylamine provides N_9 of the purine ring.

   The synthesis of phosphoribosylamine from PRPP is the **first committed (rate limiting) step** in the formation of inosine monophosphate (IMP).
3) C_4, C_5, and N_7 are next provided by the addition of amino acid **glycine** to form glycinamide ribosyl- 5-phosphate (IV). ATP is consumed in this reaction and the enzyme **phosphoribosyl glycinamide synthetase** is required.
4) A one carbon unit is next transferred to the free amino group of (IV) by an enzyme **glycinamide ribosyl-5-phosphate formyl transferase** to form formylglycinamide ribosyl-5-phosphate (V). **N^5, N^{10}-methenyl tetrahydrofolate** serves as the carrier of 1-C-unit. This reaction adds a carbon that will become C_8 of the purine ring.
5) The N_3 of the purine structure is introduced by another amination using **glutamine** and ATP to form formylglycinamidine ribosyl-5-phosphate (VI)

catalyzed by **formylglycinamidine ribosyl-5- phosphate synthetase** (VI synthetase).

6) In a reaction catalyzed by **aminoimidazole ribosyl- 5-phosphate synthetase (VII synthetase),** loss of water accompanied by ring closure forms aminoimidazole ribosyl-5-phosphate (VII).

7) Addition of CO_2 to (VII) adds the atom that will become C_6 of the purine structure. The reaction, catalyzed by **aminoimidazole ribosyl-5 phosphate carboxylase (VII carboxylase)**, requires neither ATP nor biotin and forms aminoimidazole carboxylate ribosyl-5-phosphate (VIII).

8) Condensation of **aspartate** with (VIII), catalyzed by **succinyl carboxamide ribosyl-5-phosphate synthetase (IX synthetase)**, forms aminoimidazole succinyl carboxamide ribosyl-5-phosphate (IX).

9) Liberation of the succinyl group of (IX) as fumarate, catalyzed by **adenylosuccinase**, forms aminoimidazole carboxamide ribosyl-5- phosphate (X). Reactions 8 and 9, add the atom that becomes **nitrogen-1** of the purine structure.

10) **Carbon-2** of the purine is added by **N^{10}-formyl tetrahydrofolate** to form formiminoimidazole carboxamide ribosyl-5-phosphate (XI).

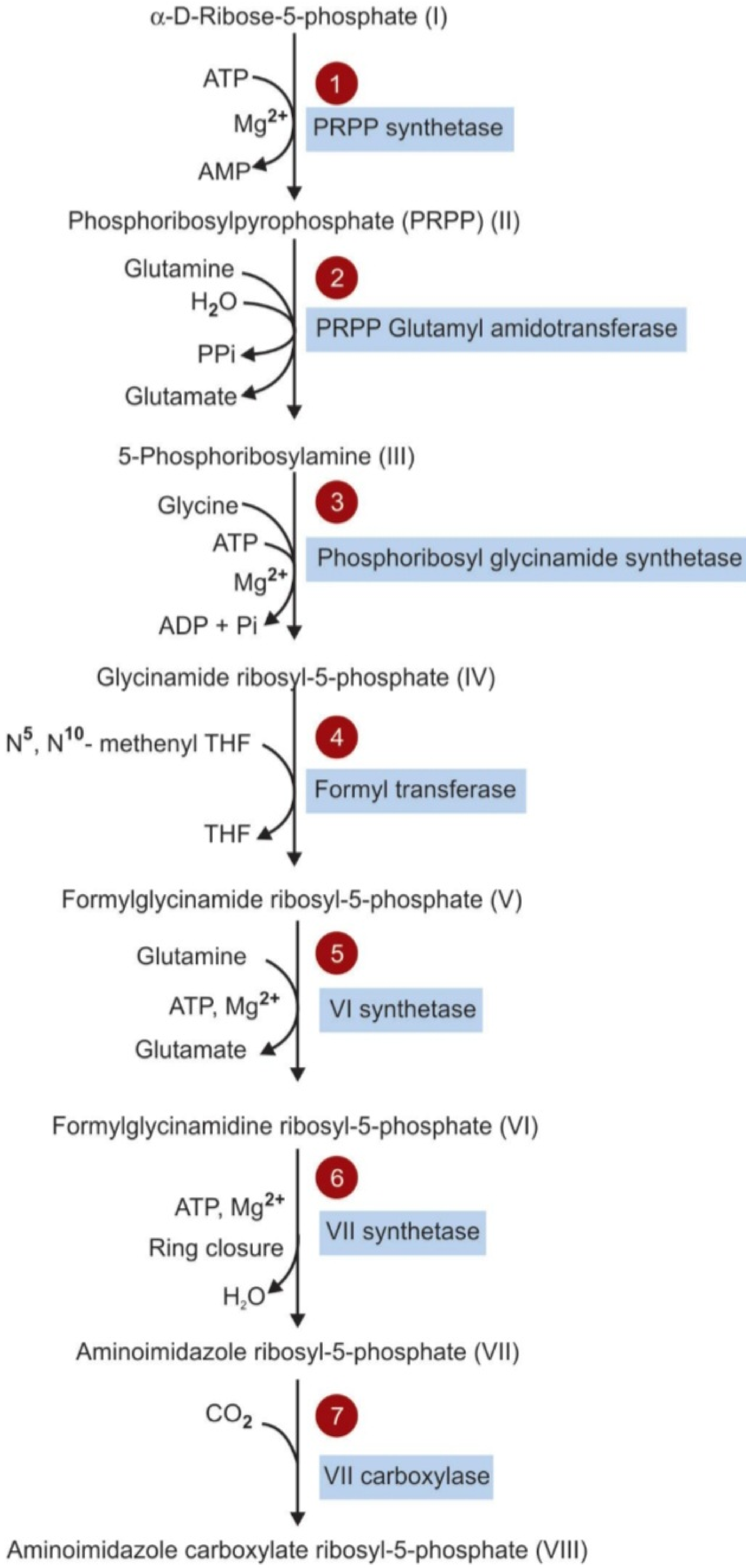
11) Ring closure of (XI) catalyzed by **IMP cyclohydrolase** forms the first purine nucleotide, inosine monophosphate, IMP (XII).

12) Addition of aspartate to IMP forms adenylosuccinate in the presence of **adenylosuccinate synthetase** and GTP.

13) **Adenylosuccinase** in turn catalyzes the removal of fumarate to yield AMP.

14) To produce GMP, the IMP must first be oxidized to xanthosine monophosphate (XMP) by NAD+ linked enzyme **IMP dehydrogenase**.

15) XMP then accepts an amino group from glutamine in the presence of ATP and the enzyme **guanosine phosphate synthetase**.


α-D-Ribose-5-phosphate (I)
ATP
Mg^{2+}
AMP
1
PRPP synthetase
Phosphoribosylpyrophosphate (PRPP) (II)
Glutamine
H_2O
PPi
Glutamate
2
PRPP Glutamyl amidotransferase
5-Phosphoribosylamine (III)
Glycine
ATP
Mg^{2+}
ADP + Pi
3
Phosphoribosyl glycinamide synthetase
Glycinamide ribosyl-5-phosphate (IV)
N^5, N^{10}- methenyl THF
THF
4
Formyl transferase
Formylglycinamide ribosyl-5-phosphate (V)
Glutamine
ATP, Mg^{2+}
Glutamate
5
VI synthetase
Formylglycinamidine ribosyl-5-phosphate (VI)
ATP, Mg^{2+}
Ring closure
H_2O
6
VII synthetase
Aminoimidazole ribosyl-5-phosphate (VII)
CO_2
7
VII carboxylase
Aminoimidazole carboxylate ribosyl-5-phosphate (VIII)

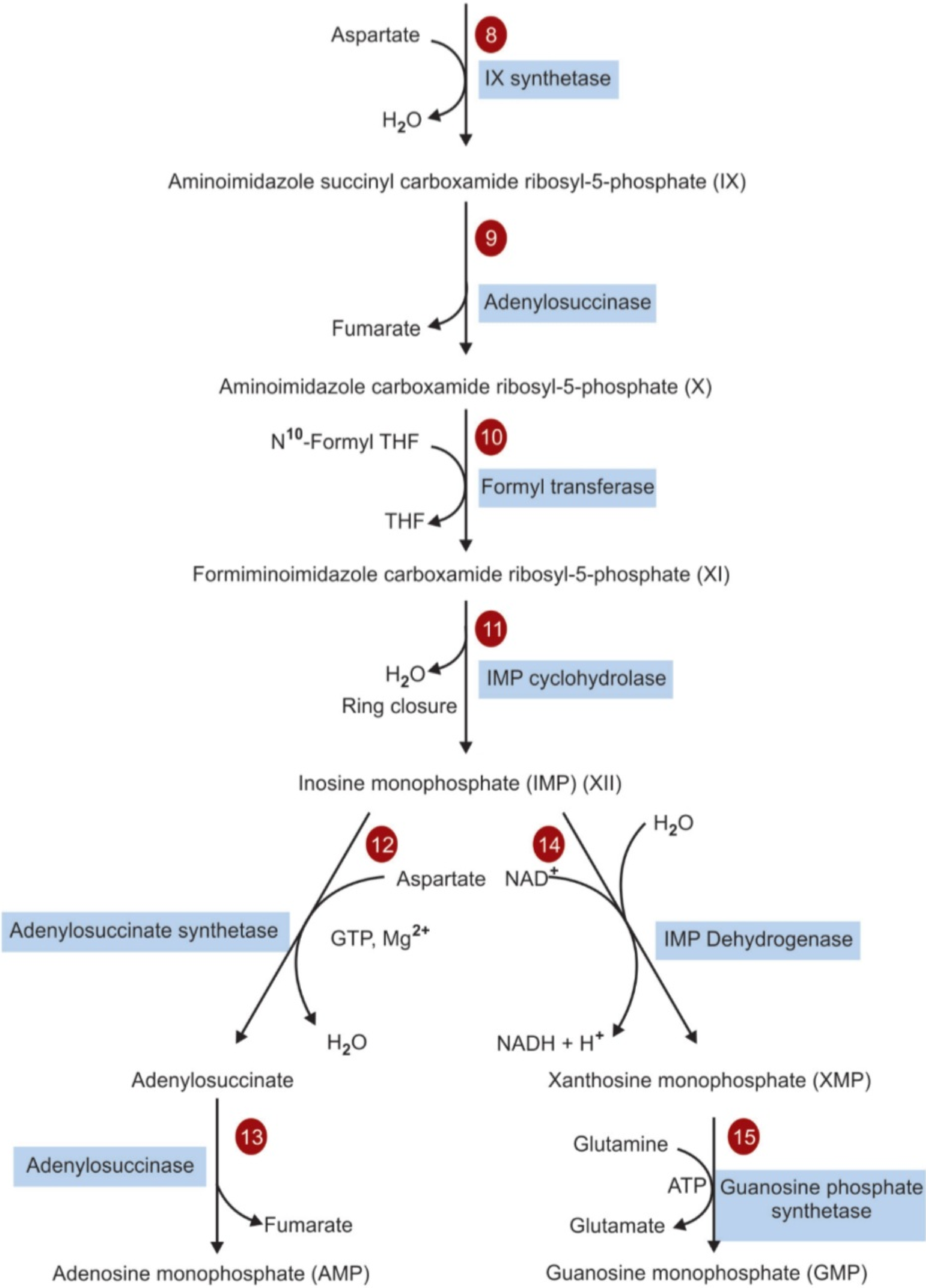


Figure 19.2: *De novo* pathway for synthesis of purine nucleotides

REGULATION OF DE NOVO SYNTHESIS OF PURINE NUCLEOTIDE (FIGURE 19.3):

- The synthesis of purine nucleotides is controlled by:
  - Concentration of PRPP
  - Feedback regulation at several sites.
- Increased concentration of PRPP stimulates the purine nucleotides synthesis. Concentration of PRPP depends on:
  - Availability of ribose-5-phosphate
  - On the activity of PRPP synthase.
- Three major feedback mechanisms regulate the overall rate of de novo purine nucleotide synthesis.
  1) The step leading to formation of PRPP. This reaction is catalyzed by an allosteric enzyme **PRPP synthetase**, which is feedback inhibited by purine nucleotides, AMP and GMP (Figure 19.3).
  2) The committed step in purine nucleotide biosynthesis is the conversion of PRPP into phosphoribosylamine by **PRPP glutamyl-amido transferase** which is feedback inhibited by AMP and GMP.
  3) AMP and GMP feedback regulate their formation from IMP. AMP feedback regulates **adenylo-succinate synthase** and GMP feedback regulates **IMP dehydrogenase.**

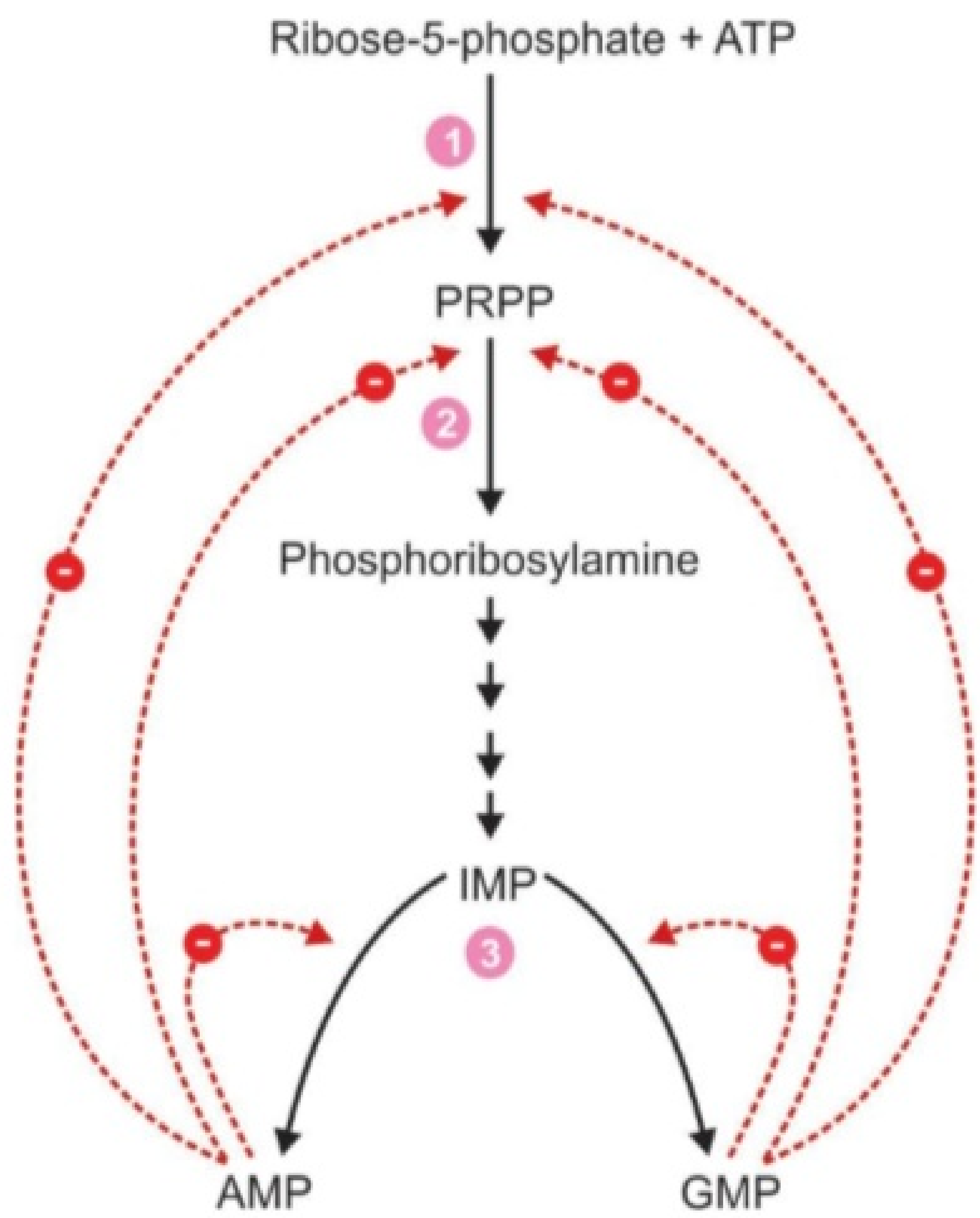


Figure 19.3: Regulation of *de novo* synthesis of purine nucleotides

2) SYNTHESIS OF PURINE NUCLEOTIDES BY SALVAGE PATHWAY:

- The pathway involved in the conversion of free purines to nucleotides is called **salvage pathway** (Salvage means property saved from loss).
- Free purine bases (adenine, guanine and hypoxanthine) are formed in cells during the metabolic degradation of nucleic acids and nucleotides. However, free purines are salvaged and used over again to remake purine nucleotides. This occurs by a pathway that is quite different from the de novo biosynthesis of purine nucleotides described earlier, in which the purine ring system is assembled step by step on ribose-5-phosphate in a long series of reactions.
- The salvage pathway is much simpler and requires far less energy than does de novo synthesis. It consists of a single reaction.

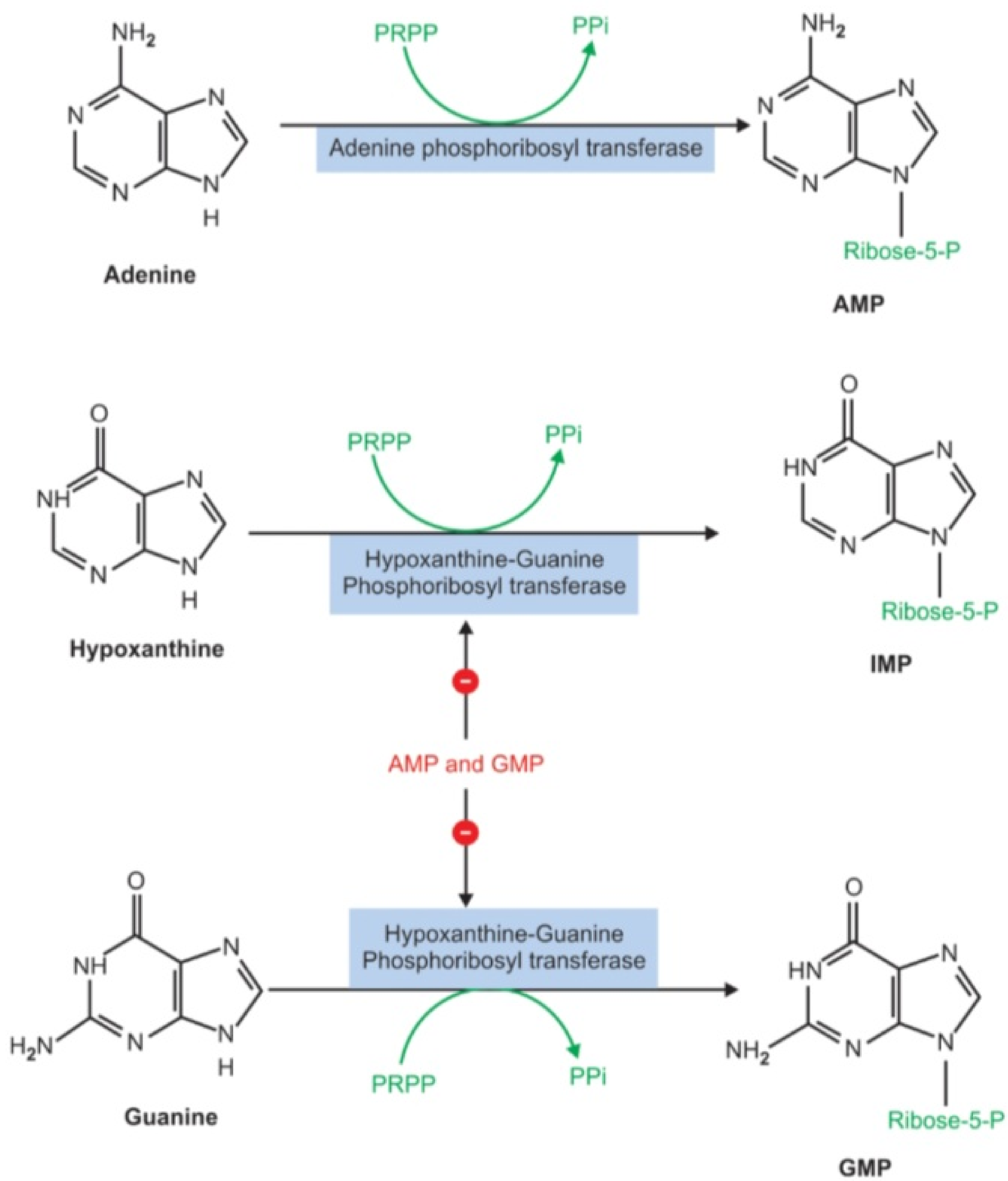


Figure 19.4: Salvage pathway of purine nucleotide synthesis

SALVAGE REACTION (FIGURE 19.4):

In salvage reaction ribose phosphate moiety of PRPP is transferred to the purine to form the corresponding nucleotide. There are two salvage enzymes with different specificities as follows:

- Adenine phosphoribosyl transferase (APRTase) catalyzes the formation of adenine nucleotide (AMP) from adenine.
- Whereas, **Hypoxanthine Guanine phosphoribosyl transferase (HGPRTase),** catalyzes the formation of ionosine (IMP) from hypoxanthine and guanine nucleotide (GMP) from guanine **(Figure 19.4).**

CATABOLISM OF PURINE NUCLEOTIDES:

The end product of purines (adenine and guanine) in humans is sparingly soluble uric acid (Figure 19.5).

- Purine nucleotides (AMP and GMP) are degraded by a pathway, in which the phosphate group is removed by the action of **nucleotidase**; to yield the nucleoside, adenosine or guanosine.
- Adenosine is then deaminated to **inosine** by **adenosine deaminase**.
- Inosine is then hydrolyzed by **purine nucleoside phosphorylase** to yield its purine base **hypoxanthine** and ribose-1-phosphate.
- Hypoxanthine is oxidized successively to **xanthine** and then **uric acid**, by **xanthine oxidase**, a molybdenum and iron containing flavoprotein. In this reaction, molecular oxygen is reduced to H_2O_2, which is decomposed to H_2O and O_2, by catalase.
- Guanosine is cleaved to **guanine** and ribose-1- phosphate by **phosphorylase** enzyme.
- Guanine undergoes hydrolytic removal of its amino group by **guanase** to yield **xanthine**, which is converted to **uric acid** by **xanthine oxidase**.

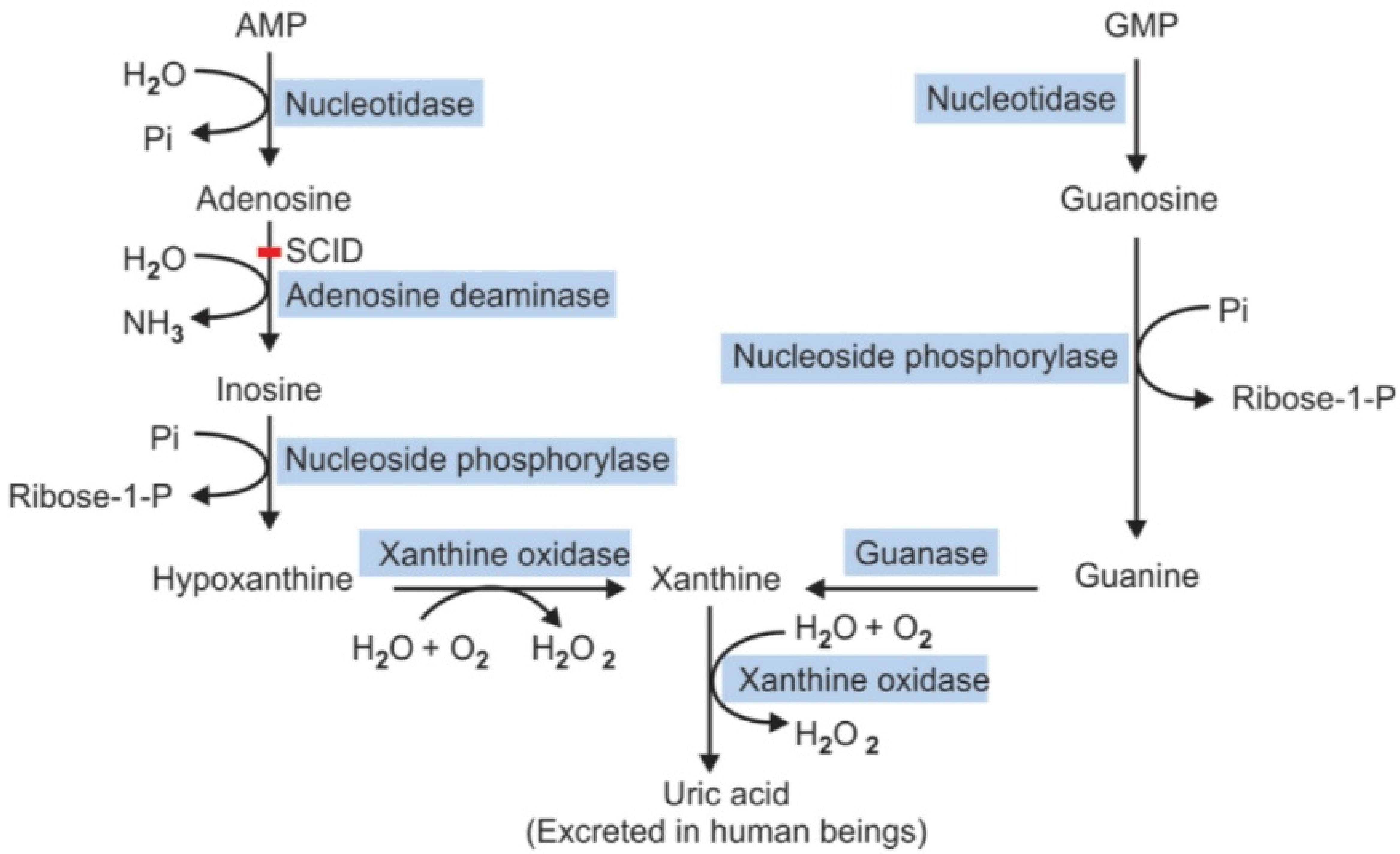


Figure 19.5: Catabolism of purine nucleotides